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SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING																																					
Program Name: B. Tech		Assignment Type: Lab	Academic Year: 2025-2026																																				
Course Coordinator Name		Dr. Rishabh Mittal																																					
Instructor(s) Name		<table border="1"> <tr><td>Mr. S Naresh Kumar</td><td></td></tr> <tr><td>Ms. B. Swathi</td><td></td></tr> <tr><td>Dr. Sasanko Shekhar Gantayat</td><td></td></tr> <tr><td>Mr. Md Sallauddin</td><td></td></tr> <tr><td>Dr. Mathivanan</td><td></td></tr> <tr><td>Mr. Y Srikanth</td><td></td></tr> <tr><td>Ms. N Shilpa</td><td></td></tr> <tr><td>Dr. Rishabh Mittal (Coordinator)</td><td></td></tr> <tr><td>Dr. R. Prashant Kumar</td><td></td></tr> <tr><td>Mr. Ankushavali MD</td><td></td></tr> <tr><td>Mr. B Viswanath</td><td></td></tr> <tr><td>Ms. Sujitha Reddy</td><td></td></tr> <tr><td>Ms. A. Anitha</td><td></td></tr> <tr><td>Ms. M. Madhuri</td><td></td></tr> <tr><td>Ms. Katherashala Swetha</td><td></td></tr> <tr><td>Ms. Velpula sumalatha</td><td></td></tr> <tr><td>Mr. Bingi Raju</td><td></td></tr> <tr><td>Mr. G. Kranthi</td><td></td></tr> </table>		Mr. S Naresh Kumar		Ms. B. Swathi		Dr. Sasanko Shekhar Gantayat		Mr. Md Sallauddin		Dr. Mathivanan		Mr. Y Srikanth		Ms. N Shilpa		Dr. Rishabh Mittal (Coordinator)		Dr. R. Prashant Kumar		Mr. Ankushavali MD		Mr. B Viswanath		Ms. Sujitha Reddy		Ms. A. Anitha		Ms. M. Madhuri		Ms. Katherashala Swetha		Ms. Velpula sumalatha		Mr. Bingi Raju		Mr. G. Kranthi	
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Course Code	23CS002PC304	Course Title	AI Assisted Coding																																				
Year/Sem	III/I	Regulation	R23																																				
Date and Day of Assignment	Week 9 - Thursday	Time(s)	23CSBTB01 To 23CSBTB52																																				
Duration	2 Hours	Applicable to Batches	All Batches																																				
Assignment Number: 18.4 (Present assignment number)/24 (Total number of assignments)																																							
Q.No.	Question	Expected Time to complete																																					
1	Lab 18: API Integration: Connecting to External Services with Error Handling	Week 9																																					

	Lab Objectives • Learn how to integrate Python programs with external REST APIs. • Understand API request/response handling using requests or similar libraries. • Implement proper error handling for failed API calls (timeouts, invalid responses, rate limits). • Practice extracting and displaying meaningful information from API responses. Learning Outcome After completing this assignment, students will be able to: • Integrate external REST APIs using AI-assisted Python scripting. • Implement structured error handling for API failures and invalid inputs. • Safely parse and validate JSON responses from external services. • Develop resilient applications capable of handling real-world API limitations.	
	Task 1 – Movie Database API Integration Scenario You are building a movie recommendation tool that fetches movie details from a public Movie Database API. Task Use AI to generate a Python script that connects to a Movie API and retrieves movie information by title. Instructions • Use <code>requests</code> library to fetch movie details. • Accept movie title as user input. • Display: ◦ Title ◦ Release year ◦ Rating ◦ Plot summary • Implement error handling for: ◦ Invalid movie name ◦ API key errors	

- Network timeout
- Empty API response

Expected Output

- A working Python script that fetches movie details.
- Graceful error messages instead of crashes.
- Clean, readable output format.

```

1 import os
2 import requests
3
4 API_URL = "https://www.omdbapi.com/"
5
6 def fetch_movie_details(title, api_key):
7     params = {"t": title, "apikey": api_key}
8
9     try:
10         response = requests.get(API_URL, params=params, timeout=8)
11     except requests.exceptions.Timeout:
12         return None, "Network timeout: The request took too long. Please try again."
13     except requests.exceptions.ConnectionError:
14         return None, "Network error: Unable to connect to the movie service."
15     except requests.exceptions.RequestException as e:
16         return None, f"Request error: {e}"
17
18 if not response.text or not response.text.strip():
19     return None, "Empty API response: No data received from the server."
20
21 try:
22     data = response.json()
23 except ValueError:
24     return None, "Invalid API response: Could not parse server response."

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

PS C:\Users\shash\student_conversion> c:\cd 'c:\Users\shash\student_conversion'; & 'c:\Users\shash\anaconda3\envs\shashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundle\libs\debugpy\launcher' '58016' '-c' 'c:\Users\shash\student_conversion\AAC A 18.4.py'
Enter your OMDb API key: 26ee90ca
HTTP error: 401 Client Error: Unauthorized for url: https://www.omdbapi.com/?t=Dune&apikey=26ee90ca
PS C:\Users\shash\student_conversion> c:\cd 'c:\Users\shash\student_conversion'; & 'c:\Users\shash\anaconda3\envs\shashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundle\libs\debugpy\launcher' '58016' '-c' 'c:\Users\shash\student_conversion\AAC A 18.4.py'
Enter your OMDb API key: 26ee90ca
Enter movie title: Dune
API key error: Invalid API key!

```

```

6 def fetch_movie_details(title, api_key):
7     ..
8
9     data = response.json()
10 except ValueError:
11     return None, "Invalid API response: Could not parse server response."
12
13 if response.status_code == 401:
14     error_message = data.get("Error", "Unauthorized API access.") if isinstance(data, dict) else "Unauthorized API error."
15     return None, f"API key error: {error_message}"
16
17 if response.status_code >= 400:
18     error_message = data.get("Error", "Unknown API error.") if isinstance(data, dict) else "Unknown API error."
19     return None, f"API error: {error_message}"
20
21 if not data:
22     return None, "Empty API response: No data found."
23
24 if data.get("Response") == "False":
25     error_message = data.get("Error", "Unknown API error.")
26     lowered = error_message.lower()
27
28 if "api key" in lowered:
29     return None, f"API key error: {error_message}"
30
31 if "movie not found" in lowered:

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

PS C:\Users\shash\student_conversion> c:\cd 'c:\Users\shash\student_conversion'; & 'c:\Users\shash\anaconda3\envs\shashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundle\libs\debugpy\launcher' '58016' '-c' 'c:\Users\shash\student_conversion\AAC A 18.4.py'
Enter your OMDb API key: 26ee90ca
HTTP error: 401 Client Error: Unauthorized for url: https://www.omdbapi.com/?t=Dune&apikey=26ee90ca
PS C:\Users\shash\student_conversion> c:\cd 'c:\Users\shash\student_conversion'; & 'c:\Users\shash\anaconda3\envs\shashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundle\libs\debugpy\launcher' '58016' '-c' 'c:\Users\shash\student_conversion\AAC A 18.4.py'
Enter your OMDb API key: 26ee90ca
Enter movie title: Dune
API key error: Invalid API key!

```

```
AAC A 18.4.py 1 X
C: > Users > shash > student_conversion > AAC A 18.4.py > ...
6 def fetch_movie_details(title, api_key):
40
41     if "api key" in lowered:
42         return None, f"API key error: {error_message}"
43     if "movie not found" in lowered:
44         return None, "Invalid movie name: No movie found with that title."
45     return None, f"API error: {error_message}"
46
47     title_value = data.get("Title", "N/A")
48     year_value = data.get("Year", "N/A")
49     rating_value = data.get("imdbRating", "N/A")
50     plot_value = data.get("Plot", "N/A")
51
52     if all(v in ("", "N/A", None) for v in [title_value, year_value, rating_value, plot_value]):
53         return None, "Empty API response: Movie data is missing."
54
55     return {
56         "Title": title_value,
57         "Release Year": year_value,
58         "Rating": rating_value,
59         "Plot Summary": plot_value
60     }, None
61
62 def main():
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\shash\student_conversion> c:: cd 'c:\Users\shash\student_conversion'; & 'c:\Users\shash\anaconda3\envs\Sh
ashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\lau
ncher' '58016' '--' 'c:\Users\shash\student_conversion\AAC A 18.4.py'
Enter your OMDb API key: 26ee90ca
HTTP error: 401 Client Error: Unauthorized for url: https://www.omdbapi.com/?t=Dune&apikey=26ee90ca
PS C:\Users\shash\student_conversion> c:: cd 'c:\Users\shash\student_conversion'; & 'c:\Users\shash\anaconda3\envs\Sh
ashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\lau
ncher' '52859' '--' 'c:\Users\shash\student_conversion\AAC A 18.4.py'
Enter your OMDb API key: 26ee90ca
Enter movie title: Dune
API key error: Invalid API key!
PS C:\Users\shash\student_conversion>
```

AAC A 18.4.py 1 X

C: > Users > shash > student_conversion > AAC A 18.4.py > ...

```
62 def main():
69     return
70
71     movie_title = input("Enter movie title: ").strip()
72     if not movie_title:
73         print("Invalid movie name: Title cannot be empty.")
74         return
75
76     movie, error = fetch_movie_details(movie_title, api_key)
77
78     if error:
79         print(error)
80         return
81
82     print("\nMovie Details")
83     print("-" * 40)
84     print(f"Title      : {movie['Title']}")
85     print(f"Release Year : {movie['Release Year']}")
86     print(f"Rating     : {movie['Rating']}")
87     print(f"Plot Summary : {movie['Plot Summary']}")
88
89 if __name__ == "__main__":
90     main()
```

PROBLEMS 1

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

```
PS C:\Users\shash\student_conversion> c::; cd 'c:\Users\shash\student_conversion\
ashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025
ncher' '58016' '--' 'c:\Users\shash\student_conversion\AAC A 18.4.py'
Enter your OMDb API key: 26ee90ca
```

```
HTTP error: 401 Client Error: Unauthorized for url: https://www.omdbapi.com/?t=
```

```
● PS C:\Users\shash\student_conversion> c::; cd 'c:\Users\shash\student_conversion\
ashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025
ncher' '52859' '--' 'c:\Users\shash\student_conversion\AAC A 18.4.py'
```

```
Enter your OMDb API key: 26ee90ca
```

```
Enter movie title: Dune
```

```
API key error: Invalid API key!
```

Task 2 – Public Transport Schedule API

Scenario

A city transport app needs to fetch bus/train arrival timings from a transport API.

Task

Use AI assistance to integrate a transport schedule API.

Instructions

- Fetch arrival times for a given station/stop ID.
- Validate user input.
- Handle errors such as:
 - Invalid stop ID
 - Server unavailable
 - Timeout errors
- Display next 3 arrival times in formatted output.
- Implement retry logic (minimum 1 retry attempt).

Expected Output

- Python script showing upcoming arrivals.
- Retry mechanism working properly.
- Proper exception handling using try-except blocks.

AAAC 18.4py X

C:\Users\shash> shash > student_conversion > AAAC 18.4py > ...

```
1 from datetime import datetime
2 import requests
3
4 API_URL_TEMPLATE = "https://api.tfl.gov.uk/stoppoint/{stop_id}/arrivals"
5 REQUEST_TIMEOUT_SECONDS = 8
6 MAX_RETRIES = 1
7
8
9 def is_valid_stop_id(stop_id):
10     if not stop_id:
11         return False
12     return stop_id.replace("_", "").replace("-", "").isalnum()
13
14
15 def fetch_arrivals(stop_id, retries=MAX_RETRIES):
16     url = API_URL_TEMPLATE.format(stop_id=stop_id)
17
18     for attempt in range(retries + 1):
19         try:
20             response = requests.get(url, timeout=REQUEST_TIMEOUT_SECONDS)
21
22             if response.status_code == 404:
23                 return None, "Invalid stop ID: Stop not found."
24
25         except requests.exceptions.ConnectionError:
26             pass
27         except requests.exceptions.Timeout:
28             pass
29         except requests.exceptions.HTTPError:
30             pass
31         except Exception:
32             pass
33
34     if response is None:
35         return None, "Server unavailable or connection issues."
36
37     if response.json() is None:
38         return None, "Empty API response / bad JSON."
39
40     arrivals = response.json().get('arrivals', [])
41
42     if not arrivals:
43         return None, "No arrivals found for the given stop ID."
44
45     return arrivals, None
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\shash\student_conversion> & 'c:\Users\shash\anaconda3\envs\shashidhar\python.exe' 'c:\Users\shash\vscode\conversion\AAAC 18.4.py'

Enter station/stop ID: 400008660H

Next 3 Arrivals

1. Line: 88

Destination : Parliament Hill Fields

ETA : 1 min

Arrival Time: 2026-03-26 08:17:53

2. Line: 214

CLAUDE CODE CHAT

MOVIE RECOMMENDATION TOOL API INTEGRATION SCRIPT

with robust error handling.

Finished with 2 steps

Implemented task 2 in AAAC 18.4py with all required behavior:

1. Fetches arrivals for a user-provided stop ID using a public transport API (TfL).

2. Validates stop ID input before API call.

3. Handles:

4. Invalid stop ID (404)

5. Server unavailable (5xx / connection issues)

6. Timeout errors

7. Empty API response / bad JSON

8. Includes retry logic with minimum 1 retry attempt (i.e. MAX_RETRIES + 1).

9. Displays next 3 arrivals in a clean formatted output.

How to run:

1. Use your current interpreter/environment.

2. Run AAAC 18.4py.

3. Enter a stop ID (example TfL stop IDs look like 400008660H).

If you want, I can also add a small built-in list of sample valid stop IDs so you can test quickly without searching one manually.

GPT-5.3-Code - 53s

```
AAC A 18.4.py X
C:\Users\shash>shash>student_conversion>AAC A 18.4.py>...
15 def fetch_arrivals(stop_id, retries=MAX_RETRIES):
26     if response.status_code >= 500:
27         if attempt < retries:
28             continue
29         return None, "Server unavailable: Transport API is currently down."
30     response.raise_for_status()
31
32     if not response.text or not response.text.strip():
33         return None, "Empty API response: No data received from server."
34
35     data = response.json()
36
37     if not isinstance(data, list):
38         return None, "Invalid API response format."
39
40     if len(data) == 0:
41         return [], None
42
43     return data, None
44
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shash\student_conversion> & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c:\Users\shash\.vscode
2. Line: 214
Destination : Highgate Village
ETA : 6 min
Arrival Time: 2026-03-26 08:22:30
-----
3. Line: 88
Destination : Parliament Hill Fields
ETA : 12 min
Arrival Time: 2026-03-26 08:29:00
-----
1 Hidden T...
```

```
AAC A 18.4.py X
C:\Users\shash>shash>student_conversion>AAC A 18.4.py>...
15 def fetch_arrivals(stop_id, retries=MAX_RETRIES):
44
45     except requests.exceptions.Timeout:
46         if attempt < retries:
47             continue
48         return None, "Timeout error: Request timed out after retry."
49     except requests.exceptions.ConnectionError:
50         if attempt < retries:
51             continue
52         return None, "Server unavailable: Could not connect to transport service."
53     except requests.exceptions.HTTPError as err:
54         return None, f"HTTP error: {err}"
55     except requests.exceptions.RequestException as err:
56         return None, f"Request error: {err}"
57     except ValueError:
58         return None, "Invalid API response: Could not parse JSON."
59
60     return None, "Unexpected error during retry handling."
61
62
63 def format_time(iso_timestamp):
64     try:
65         parsed = datetime.fromisoformat(iso_timestamp.replace("Z", "+00:00"))
66         return parsed.strftime("%Y-%m-%d %H:%M:%S")
67
68     except Exception as e:
69         return None, f"Error formatting time: {e}"
70
71     return None, "Unexpected error during time formatting."
72
73
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shash\student_conversion> & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c:\Users\shash\.vscode
2. Line: 214
Destination : Highgate Village
ETA : 6 min
Arrival Time: 2026-03-26 08:22:30
-----
3. Line: 88
Destination : Parliament Hill Fields
ETA : 12 min
Arrival Time: 2026-03-26 08:29:00
-----
1 Hidden T...
```

AAC A 18.4.py X

C:\Users\shash> student_conversion > AAC A 18.4.py > ...

```
63 def format_time(iso_timestamp):
64     return "Unknown"
65
66
67
68
69
70
71 def get_next_three_arrivals(arrivals):
72     ordered = sorted(arrivals, key=lambda x: x.get("timeToStation", float("inf")))
73     return ordered[:3]
74
75
76
77 def main():
78     stop_id = input("Enter station/stop ID: ").strip()
79
80     if not is_valid_stop_id(stop_id):
81         print("Invalid stop ID: Use letters, numbers, '-' or '_' only.")
82         return
83
84     arrivals, error = fetch_arrivals(stop_id)
85
86     if error:
87         print(error)
88         return
89
90     if not arrivals:
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\shash\student_conversion> & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c:\Users\shash\.vscode

```
2. Line: 214
Destination : Highgate Village
ETA         : 6 min
Arrival Time: 2026-03-26 08:22:30
-----
```

```
3. Line: 88
Destination : Parliament Hill Fields
ETA         : 12 min
Arrival Time: 2026-03-26 08:29:00
-----
```


AAC A 18.4.py X

C: > Users > shash > student_conversion > AAC A 18.4.py > ...

```
76 def main():
89     if not arrivals:
90         print("No upcoming arrivals found for this stop.")
91         return
92
93     next_arrivals = get_next_three_arrivals(arrivals)
94
95     print("\nNext 3 Arrivals")
96     print("-" * 60)
97     for index, item in enumerate(next_arrivals, start=1):
98         line_name = item.get("lineName", "N/A")
99         destination = item.get("destinationName", "N/A")
100         expected = format_time(item.get("expectedArrival", ""))
101         minutes = round(item.get("timeToStation", 0) / 60)
102         print(f"{index}. Line: {line_name}")
103         print(f"    Destination : {destination}")
104         print(f"    ETA           : {minutes} min")
105         print(f"    Arrival Time: {expected}")
106         print("-" * 60)
107
108 if __name__ == "__main__":
109     main()
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\shash\student_conversion> & 'c:\Users\shash\anaconda3\envs\Shashidhar\python

```
2. Line: 214
   Destination : Highgate Village
   ETA         : 6 min
   Arrival Time: 2026-03-26 08:22:30
-----
3. Line: 88
   Destination : Parliament Hill Fields
   ETA         : 12 min
   Arrival Time: 2026-03-26 08:29:00
-----
```

Task 3 – Stock Market Data API

Scenario

You are developing a finance dashboard that retrieves live stock prices.

Task

Use AI to generate a Python script that fetches stock data from a financial API.

Instructions

- Accept stock symbol as input.

- Retrieve:
 - Current price
 - Day high
 - Day low
- Handle:
 - Invalid stock symbol
 - API rate limits
 - Missing fields in JSON response
- Validate and safely parse JSON response.

Expected Output

- Script fetches and displays stock information.
- No unhandled exceptions.
- Clear error messages for invalid symbols or rate limits.

```

AAC A 18.4.py X
C:\Users\shash\student_conversion > AAC A 18.4.py > ...
1 import requests
2
3 API_URL = "https://www.alphavantage.co/query"
4 REQUEST_TIMEOUT_SECONDS = 10
5
6
7 def is_valid_symbol(symbol):
8     if not symbol:
9         return False
10    return symbol.replace("-", "").replace(".", "").isalnum()
11
12
13 def to_float(value):
14     if value is None:
15         return None
16     try:
17         return float(str(value).strip())
18     except (TypeError, ValueError):
19         return None
20
21
22 def fetch_stock_data(symbol, api_key):
23     params = {
24         "function": "GLOBAL_QUOTE"
25     }
26
27     response = requests.get(API_URL, params=params, timeout=REQUEST_TIMEOUT_SECONDS)
28     data = response.json()
29
30     if "Error" in data:
31         print(f"Error: {data['Error']}")
32         return None
33
34     quote = data["Global Quote"]
35
36     symbol_info = {
37         "Symbol": symbol,
38         "Current Price": quote["05. price"],
39         "Day High": quote["04. high"],
40         "Day Low": quote["03. low"]
41     }
42
43     print("Stock Information")
44     print("-----")
45     for key, value in symbol_info.items():
46         print(f"{key} : {value}")
47
48 if __name__ == "__main__":
49     symbol = input("Enter Alpha Vantage API key (or type demo): ")
50     api_key = "demo"
51
52     if not is_valid_symbol(symbol):
53         print("Invalid stock symbol")
54     else:
55         fetch_stock_data(symbol, api_key)
56
57 PS C:\Users\shash\student_conversion> cd 'c:\Users\shash\student_conversion'; & 'c:\Users\shash\anaconda3\envs\shashidhar\python.exe' 'c:\Users\shash\vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bin\debugpy\launcher' '64662' '-.' 'c:\Users\shash\student_conversion\AAC A 18.4.py'
Enter Alpha Vantage API key (or type demo): demo
Stock Information
-----
Symbol : IBM
Current Price: 241.39
Day High : 246.19
Day Low : 238.00
  
```

MOVIE RECOMMENDATION TOOL API INTEGRATION SCRIPT

5. Day high
6. Day low
7. Handles:
8. Invalid stock symbol
9. API rate limits
10. Missing fields in JSON response
11. Timeout, connection, and HTTP request errors
12. Validates and safely parses JSON using strict type checks and safe float conversion.
13. Prevents unhandled exceptions by returning clear error messages.

What I verified:

1. Static diagnostics show no errors in AAC A 18.4.py.
2. Runtime test through the non-interactive terminal failed at input prompt with EOF, which is expected for input-based scripts in that terminal mode.

How to run interactively:

1. Start the file in VS Code Run/Debug as you did before.
2. Enter API key when prompted (you can try demo).
3. Enter a stock symbol such as IBM.

If you want, I can convert it to support command-line arguments so it can run in non-interactive terminals too.

AAC A 18.4.py

Describe what to build

+ Agent Auto

AAC A 18.4.py X

C: > Users > shash > student_conversion > AAC A 18.4.py > ...

```
22 def fetch_stock_data(symbol, api_key):
23     """Fetch stock data from Alpha Vantage API"""
24     symbol = symbol.upper()
25     "symbol": symbol,
26     "apikey": api_key,
27 }
28
29 try:
30     response = requests.get(API_URL, params=params, timeout=REQUEST_TIMEOUT_SECONDS)
31     response.raise_for_status()
32 except requests.exceptions.Timeout:
33     return None, "Timeout error: Request to stock API timed out."
34 except requests.exceptions.ConnectionError:
35     return None, "Server unavailable: Could not connect to stock API."
36 except requests.exceptions.HTTPError as err:
37     return None, f"HTTP error: {err}"
38 except requests.exceptions.RequestException as err:
39     return None, f"Request error: {err}"
40
41 if not response.text or not response.text.strip():
42     return None, "Empty API response: No data received from server."
43
44 try:
45     data = response.json()
46 except ValueError:
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\shash\student_conversion> c::; cd 'c:\Users\shash\student_conversion'; & 'c:\Users\shash\anashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libncher' '64962' '--' 'c:\Users\shash\student_conversion\AAC A 18.4.py'

Enter Alpha Vantage API key (or type demo): demo

Stock Information

Symbol : IBM
Current Price: 241.39
Day High : 246.19
Day Low : 238.00

PS C:\Users\shash\student_conversion> |

AAC A 18.4.py X

C: > Users > shash > student_conversion > AAC A 18.4.py > ...

```
22 def fetch_stock_data(symbol, api_key):
46     except ValueError:
47         return None, "Invalid JSON response: Could not parse API response."
48
49     if not isinstance(data, dict):
50         return None, "Invalid JSON response: Expected an object."
51
52     if "Note" in data:
53         return None, "API rate limit reached: Please wait and try again."
54
55     if "Error Message" in data:
56         return None, "Invalid stock symbol: No matching symbol found."
57
58     quote = data.get("Global Quote")
59     if not isinstance(quote, dict) or not quote:
60         return None, "Invalid stock symbol: No quote data returned."
61
62     symbol_value = quote.get("01. symbol")
63     price_value = to_float(quote.get("05. price"))
64     high_value = to_float(quote.get("03. high"))
65     low_value = to_float(quote.get("04. low"))
66
67     missing = []
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shash\student_conversion> c:: cd 'c:\Users\shash\student_conversion'; & 'c:\U
ashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-
ncher' '64962' '--' 'c:\Users\shash\student_conversion\AAC A 18.4.py'
```

Enter Alpha Vantage API key (or type demo): demo

Stock Information

```
-----
Symbol      : IBM
Current Price: 241.39
Day High    : 246.19
Day Low     : 238.00
```

PS C:\Users\shash\student_conversion>

```
AAC A 18.4.py X
C: > Users > shash > student_conversion > AAC A 18.4.py > ...
88 def main():
92     if not api_key:
93         print("API key is required.")
94         return
95
96     if not is_valid_symbol(symbol):
97         print("Invalid stock symbol: Use letters, numbers, '.', or '-'.")
98         return
99
100    stock, error = fetch_stock_data(symbol, api_key)
101
102    if error:
103        print(error)
104        return
105
106    print("\nStock Information")
107    print("-" * 40)
108    print(f"Symbol      : {stock['symbol']}")
109    print(f"Current Price: {stock['current_price']:.2f}")
110    print(f"Day High    : {stock['day_high']:.2f}")
111    print(f"Day Low     : {stock['day_low']:.2f}")
112
113    if __name__ == "__main__":
114        main()

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\shash\student_conversion> c::; cd 'c:\Users\shash\student_conversion'; & 'c:\Us
ashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x
ncher' '64962' '--' 'c:\Users\shash\student_conversion\AAC A 18.4.py'
Enter Alpha Vantage API key (or type demo): demo
Stock Information
-----
Symbol      : IBM
Current Price: 241.39
Day High    : 246.19
Day Low     : 238.00
PS C:\Users\shash\student_conversion> |
```

Task 4 – Geolocation API (IP Address Lookup)

Scenario

A cybersecurity tool needs to detect the geographical location of a user based on IP address.

Task

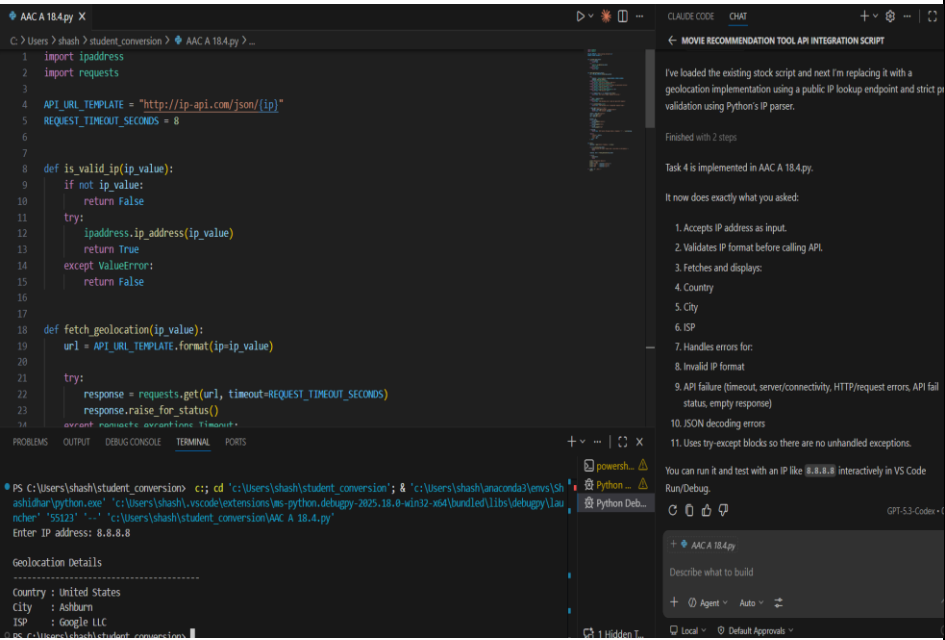
Integrate a Geolocation API using AI assistance.

Instructions

- Accept IP address as input.
- Fetch:
 - Country
 - City
 - ISP
- Validate IP format before making API call.
- Handle:
 - Invalid IP format
 - API failure
 - JSON decoding errors

Expected Output

- A script that returns geolocation details.
- Input validation implemented.
- Robust error handling included.



C: > Users > shash > student_conversion > AAC A 18.4.py > ...

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** PORTS

Geolocation Details

```
PS C:\Users\shash\student_conversion>
```

Launchpad java: Ready

AAC A 18.4.py X

C: > Users > shash > student_conversion > AAC A 18.4.py > ...

```
18 def fetch_geolocation(ip_value):
44     if data.get("status") == "fail":
45         message = data.get("message", "Lookup failed.")
46         return None, f"API failure: {message}"
47
48     country = data.get("country")
49     city = data.get("city")
50     isp = data.get("isp")
51
52     missing = []
53     if not country:
54         missing.append("country")
55     if not city:
56         missing.append("city")
57     if not isp:
58         missing.append("isp")
59
60     if missing:
61         return None, "API failure: Missing fields in response: " + ", ".join(missing)
62
63     return {
64         "country": country,
65         "city": city,
66         "isp": isp,
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
● PS C:\Users\shash\student_conversion> c:: cd 'c:\Users\shash\student_conversion'; & 'c:\Users\shash\anaco
ashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs
ncher' '55123' '-.' 'c:\Users\shash\student_conversion\AAC A 18.4.py'
Enter IP address: 8.8.8.8
```

Geolocation Details

```
-----
Country : United States
City    : Ashburn
ISP     : Google LLC
```



```
AAC A 18.4.py X
C: > Users > shash > student_conversion > AAC A 18.4.py > ...

68
69
70 def main():
71     ip_value = input("Enter IP address: ").strip()
72
73     if not is_valid_ip(ip_value):
74         print("Invalid IP format: Please enter a valid IPv4 or IPv6 address.")
75         return
76
77     location, error = fetch_geolocation(ip_value)
78
79     if error:
80         print(error)
81         return
82
83     print("\nGeolocation Details")
84     print("-" * 40)
85     print(f"Country : {location['country']}")
86     print(f"City    : {location['city']}")
87     print(f"ISP     : {location['isp']}")
88
89 if __name__ == "__main__":
90     main()

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

• PS C:\Users\shash\student_conversion> c::; cd 'c:\Users\shash\student_conversion'; & 'c:\User
ashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64
ncher' '55123' '--' 'c:\Users\shash\student_conversion\AAC A 18.4.py'
Enter IP address: 8.8.8.8

Geolocation Details
-----
Country : United States
City    : Ashburn
ISP     : Google LLC
```

Task 5 – Real-Time Application: Payment Gateway Simulation API

Scenario

You are building a test payment integration system that connects to a sandbox payment API.

Task

Use AI to generate a Python script that simulates payment processing.

Instructions

- Send POST request with:
 - Amount
 - Currency
 - Payment method
- Validate input values.
- Handle:
 - Invalid payment details
 - HTTP status errors (400, 401, 500)
 - Network interruptions
- Display transaction status clearly.
- Log failed transactions to a file.

Expected Output

- Script processes simulated payment.
- Errors handled gracefully.
- Failed requests logged properly.

AACA 18.4.py X

C:\Users\shash> shash > student_conversion > AACA 18.4.py > process_payment

```
1 import json
2 from datetime import datetime
3 import requests
4
5 API_URL = "https://httpbin.org/post"
6 REQUEST_TIMEOUT_SECONDS = 8
7 LOG_FILE = "failed_transactions.log"
8 VALID_PAYMENT_METHODS = ("card", "upi", "wallet", "netbanking")
9
10
11 def log_failed_transaction(payload, reason):
12     entry = {
13         "timestamp": datetime.datetime.now().isoformat() + "Z",
14         "payload": payload,
15         "reason": reason,
16     }
17     with open(LOG_FILE, "a", encoding="utf-8") as file:
18         file.write(json.dumps(entry) + "\n")
19
20
21 def is_valid_amount(amount_text):
22     try:
23         value = float(amount_text)
24         return value > 0
25     except ValueError:
26         return False
27
28
29 def main():
30     amount = input("Enter amount: ")
31     if not is_valid_amount(amount):
32         print("Invalid amount. Please enter a positive number.")
33         return
34     currency = input("Enter currency (3-letter code, e.g., USD): ")
35     if len(currency) != 3:
36         print("Invalid currency code. Please enter a 3-letter code.")
37         return
38     payment_method = input("Enter payment method (card/upi/wallet/netbanking): ")
39     if payment_method not in VALID_PAYMENT_METHODS:
40         print("Invalid payment method. Please choose from card, upi, wallet, or netbanking.")
41         return
42
43     payload = {
44         "amount": amount,
45         "currency": currency,
46         "payment_method": payment_method,
47     }
48
49     try:
50         response = requests.post(API_URL, json=payload, timeout=REQUEST_TIMEOUT_SECONDS)
51         response.raise_for_status()
52         print("Transaction successful!")
53     except requests.exceptions.HTTPError as http_err:
54         print(f"HTTP error occurred: {http_err}")
55     except requests.exceptions.Timeout:
56         print("Request timed out.")
57     except requests.exceptions.ConnectionError:
58         print("Connection error.")
59     except Exception as err:
60         print(f"An error occurred: {err}")
61
62     log_failed_transaction(payload, "Transaction failed")
63
64 if __name__ == "__main__":
65     main()
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\shash\student_conversion> cd 'c:\Users\shash\student_conversion'; & 'c:\Users\shash\anaconda\envs\shash\python.exe' 'c:\Users\shash\vscode\extensions\ms-python.debugpy-2025.18.8-win32-x64\bundle\libs\debugpy\launcher' '51823' '-c' 'c:\Users\shash\student_conversion\AACA 18.4.py'

Enter amount: 1000
Enter currency (3-letter code, e.g., USD): USD
Enter payment method (card/upi/wallet/netbanking): card

Transaction Status

SUCCESS
HTTP Status : 200

CLAUDE CODE CHAT

MOVIE RECOMMENDATION TOOL API INTEGRATION SCRIPT

1. amount
2. amount
3. currency
4. payment method
5. Validates inputs:
6. amount must be positive number
7. currency must be 3-letter code
8. payment method must be one of card/upi/wallet/netbanking
9. Handles errors for:
10. invalid payment details
11. HTTP status errors including 400, 401, 500
12. network interruptions (timeout/connection/request exceptions)
13. Displays transaction status clearly as SUCCESS or FAILED.
14. Logs failed transactions to file:
15. failed_transactions.log
16. includes timestamp, payload, and failure reason (JSON lines format)

Run flow:
1. Run AACA 18.4.py
2. Enter amount, currency, and payment method
3. Check failed_transactions.log for failed attempts

If you want, I can add a simple switch to intentionally simulate 400/401/500 responses so you can demo those error paths during evaluation.

AAC A 18.4.py X

C: > Users > shash > student_conversion > AAC A 18.4.py > process_payment

```
28
29 def is_valid_currency(currency):
30     if not currency:
31         return False
32     return len(currency) == 3 and currency.isalpha()
33
34
35 def is_valid_payment_method(payment_method):
36     if not payment_method:
37         return False
38     return payment_method.lower() in VALID_PAYMENT_METHODS
39
40
41 def process_payment(amount, currency, payment_method):
42     payload = {
43         "amount": amount,
44         "currency": currency.upper(),
45         "payment_method": payment_method.lower(),
46     }
47
48     try:
49         response = requests.post(API_URL, json=payload, timeout=REQUEST_TIMEOUT_SECONDS)
50     except requests.exceptions.Timeout:
51         reason = "Network interruption: Request timed out "
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shash\student_conversion> c:: cd 'c:\Users\shash\student_conversion'; & 'c:\Users\shash\anac
ashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundle\libs
ncher' '51023' '--' 'c:\Users\shash\student_conversion\AAC A 18.4.py'
Enter amount: 1000
```

```
-----
SUCCESS
HTTP Status   : 200
Amount        : 1000.00
Currency      : USD
Payment Method: card
```

PS C:\Users\shash\student_conversion>

AAC A 18.4.py X

C: > Users > shash > student_conversion > AAC A 18.4.py > process_payment

```
41 def process_payment(amount, currency, payment_method):
51     reason = "Network interruption: Request timed out."
52     log_failed_transaction(payload, reason)
53     return None, reason
54 except requests.exceptions.ConnectionError:
55     reason = "Network interruption: Unable to connect to payment gateway."
56     log_failed_transaction(payload, reason)
57     return None, reason
58 except requests.exceptions.RequestException as err:
59     reason = f"Network interruption: {err}"
60     log_failed_transaction(payload, reason)
61     return None, reason
62
63 if response.status_code == 400:
64     reason = "HTTP 400: Invalid payment details sent to gateway."
65     log_failed_transaction(payload, reason)
66     return None, reason
67 if response.status_code == 401:
68     reason = "HTTP 401: Unauthorized request to gateway."
69     log_failed_transaction(payload, reason)
70     return None, reason
71 if response.status_code == 500:
72     reason = "HTTP 500: Payment gateway server error."
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shash\student_conversion> c::; cd 'c:\Users\shash\student_conversion'; & 'c:\Users\shash\ashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundle\ncher' '51023' '-' 'c:\Users\shash\student_conversion\AAC A 18.4.py'
Enter amount: 1000
```

```
-----
SUCCESS
HTTP Status : 200
Amount : 1000.00
Currency : USD
Payment Method: card
```

AAC A 18.4.py X

C: > Users > shash > student_conversion > AAC A 18.4.py > process_payment

```
41 def process_payment(amount, currency, payment_method):
74     return None, reason
75     if response.status_code >= 400:
76         reason = f"HTTP error: {response.status_code}"
77         log_failed_transaction(payload, reason)
78         return None, reason
79
80     if not response.text or not response.text.strip():
81         reason = "API failure: Empty response from payment gateway."
82         log_failed_transaction(payload, reason)
83         return None, reason
84
85     try:
86         data = response.json()
87     except ValueError:
88         reason = "JSON decoding error: Could not parse gateway response."
89         log_failed_transaction(payload, reason)
90         return None, reason
91
92     if not isinstance(data, dict):
93         reason = "JSON decoding error: Unexpected response format."
94         log_failed_transaction(payload, reason)
95         return None, reason
96
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shash\student_conversion> c::; cd 'c:\Users\shash\student_conversion'; & 'c:\Users\shash\student_conversion\python.exe' 'c:\Users\shash\student_conversion\AAC A 18.4.py'
ncher' '51023' '--' 'c:\Users\shash\student_conversion\AAC A 18.4.py'
```

Enter amount: 1000

SUCCESS

```
HTTP Status   : 200
Amount        : 1000.00
Currency      : USD
Payment Method: card
```

PS C:\Users\shash\student_conversion> |

AAC A 18.4.py X

C: > Users > shash > student_conversion > AAC A 18.4.py > process_payment

```
41 def process_payment(amount, currency, payment_method):
96
97     return {
98         "transaction_status": "SUCCESS",
99         "http_status": response.status_code,
100         "amount": payload["amount"],
101         "currency": payload["currency"],
102         "payment_method": payload["payment_method"],
103     }, None
104
105
106 def main():
107     amount_text = input("Enter amount: ").strip()
108     currency = input("Enter currency (3-letter code, e.g., USD): ").strip().upper()
109     payment_method = input("Enter payment method (card/upi/wallet/netbanking): ").strip().lower()
110
111     if not is_valid_amount(amount_text):
112         print("Invalid payment details: Amount must be a positive number.")
113         return
114
115     if not is_valid_currency(currency):
116         print("Invalid payment details: Currency must be a 3-letter code.")
117         return
118
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\shash\student_conversion> c:: cd 'c:\Users\shash\student_conversion'; & 'c:\Users\shash\anaconda3\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundle\libs\debugpy\launcher' '51023' '---' 'c:\Users\shash\student_conversion\AAC A 18.4.py'

Enter amount: 1000

SUCCESS

HTTP Status : 200
Amount : 1000.00
Currency : USD
Payment Method: card

PS C:\Users\shash\student_conversion> |

AAC A 18.4.py X

C: > Users > shash > student_conversion > AAC A 18.4.py > process_payment

```
106 def main():
122
123     amount = float(amount_text)
124     result, error = process_payment(amount, currency, payment_method)
125
126     if error:
127         print("\nTransaction Status")
128         print("-" * 40)
129         print("FAILED")
130         print(f"Reason: {error}")
131         print(f"Failure logged to {LOG_FILE}")
132         return
133
134     print("\nTransaction Status")
135     print("-" * 40)
136     print(result["transaction_status"])
137     print(f"HTTP Status   : {result['http_status']}")
138     print(f"Amount          : {result['amount']:.2f}")
139     print(f"Currency         : {result['currency']}")
140     print(f"Payment Method: {result['payment_method']}")
141
142 if __name__ == "__main__":
143     main()
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shash\student_conversion> c::; cd 'c:\Users\shash\student_conversion';
ashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.
ncher' '51023' '--' 'c:\Users\shash\student_conversion\AAC A 18.4.py'
Enter amount: 1000
```

```
-----
SUCCESS
HTTP Status   : 200
Amount        : 1000.00
Currency      : USD
Payment Method: card
```

Note: Report should be submitted a word document for all tasks in a single document with prompts, comments & code explanation, and output and if required, screenshots